

12/10/2025 Professor Mukherjee Lab Meeting Meeting Minutes

Students submitted several questions to Professor Mukherjee regarding potential research projects:

- **Post-operative hydrocephalus:** Students asked which predictors to consider. Professor Mukherjee recommended focusing on clinical and radiographic variables.
- **Meningioma:** Professor Mukherjee referenced [ICOM](#) and advised contacting the organizers for further guidance.
- **Recraniotomy:** Key challenges were outlined, and Professor Mukherjee suggested reaching out to two professors who are leading authorities on this topic.

Professor Mukherjee emphasized that identifying a research topic is an iterative process. It begins with an understanding of the available institutional dataset and the relevant disease type. After the student articulates her interests, feasibility is assessed, including sample size and other methodological constraints. Examples discussed included:

- [QOD Tumor Registry](#): Potential predictors could include readmission, infections, and radiographic findings.
- [NCDB Database](#): This dataset has been extensively mined; however, remaining areas of interest may include:
 - Grade 1 gliomas
 - Brain metastases

Professor Mukherjee also mentioned two potential future clinical trials that could be sponsored:

- A randomized trial involving NSCLC patients and steroid use
- A multi-year, single-institution study on SSRIs

Finally, Professor Mukherjee offered support for the fMRI annotation project, which includes 40 patients.